MILITARY SPECIFICATION

MOLDING MATERIAL, POLYPROPYLENE, MODIFIED MECHANICAL USE ONLY

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for unfilled, unpigmented, modified polypropylene, either copolymer or modified homopolymer, for mechanical use only.

2. APPLICABLE DOCUMENTS

2.1 Publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D256	- Impact Resistance of Plastics and Electrical Insulating Materials
ASTM D570	- Water Absorption of Plastics
ASTM D618	- Conditioning Plastics and Electrical Insulating Materials for Testing
ASTM D621	- Deformation of Plastics Under Load
ASTM D638	- Tensile Properties of Plastics
ASTM D746	- Test for Brittleness Temperature of Plastics and Elastomers by Impact
ASTM D792	- Specific Gravity and Density of Plastics
ASTM D1238	- Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D1505	- Density of Plastics by the Density-Gradient Technique

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Air Force Wright Aeronautical Laboratories, MLSA, Wright-Patterson Air Force Base, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9330

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The material shall be unfilled, modified polypropylene homopolymer or polypropylene copolymers so formulated as to meet the requirements of this specification.

3.1.1 Color. Unless otherwise specified by the procuring activity, the material shall contain no color pigment.

3.1.2 Form. The material shall be furnished in the form of granules or pellets.

3.2 Property value. The values obtained for each set of specimens shall meet the required values specified in Table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Responsibility for documentation. Unless otherwise specified in the contract or purchase order, the fabrication supplier, or if fabrication is subcontracted, the prime contractor shall establish and maintain a record of all certificates of inspection. These certificates of inspection shall contain the average of the test value determined, the standard deviation of that average and the number of replicates used to calculate the standard deviation. In the event that four or less data points are determined, each value and the average shall be reported.

4.3 Quality conformance inspection.

4.3.1 Batch inspection of product. Batch inspection of product shall consist of a single inspection per batch for impact, water absorption, deformation under load and tensile strength. Samples shall be taken at the beginning, middle and end of batch run and combined for molding the respective test specimens.

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TABLE I. Property Values.

Property	Value Required	Method	Remarks
Impact Strength, Minimum ft lb/in of Notch, 23°C	3.0	D256	Method A, 1/8 inch wide
Water Absorption, Maximum Percent, 23°C	0.03	D570	None
Deformation Under Load Maximum, Percent, 50°C, @ 1000 psi (70 kg/cu^2)	6.0	D621	Method A
Tensile Yield Strength, Minimum, psi, 23°C	2900	D638	Type I specimen, 2 inches per minute
Brittleness Temperature, Maximum, °C	-20	D746	
Density Range, 23°C	.890 to .915	D792 or D1505	Method B (D792)
Flow Rate, Grams in 10 Minutes	1.4 to 2.0	D1238	Condition L

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4.3.2 Periodic inspection of product. Periodic inspection of product shall consist of three inspections per batch at the beginning, middle and end of the batch run for brittleness temperature, density and flow ranges. Samples taken shall not be combined and shall be identified as to what portion of the batch run they represent.

4.4 Inspection batch. A bulk batch shall consist of a quantity of a homogenious mixture of material offered for acceptance but shall not exceed the quantity contained in one bulk hopper car. A bulk lot may represent several bulk truckloads and many 1000 pound gaylords, but shall not exceed 170,000 pounds.

5. PACKAGING -- Not Applicable.

6. NOTES

6.1 Intended use. The modified polypropylene covered by this specification is intended for injection molding of tubes for carrying 30mm ammunition in the ammunition loading system for the GAU-8 gun in the A-10 aircraft and the GPU-80 pod on various other aircraft.

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